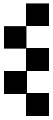
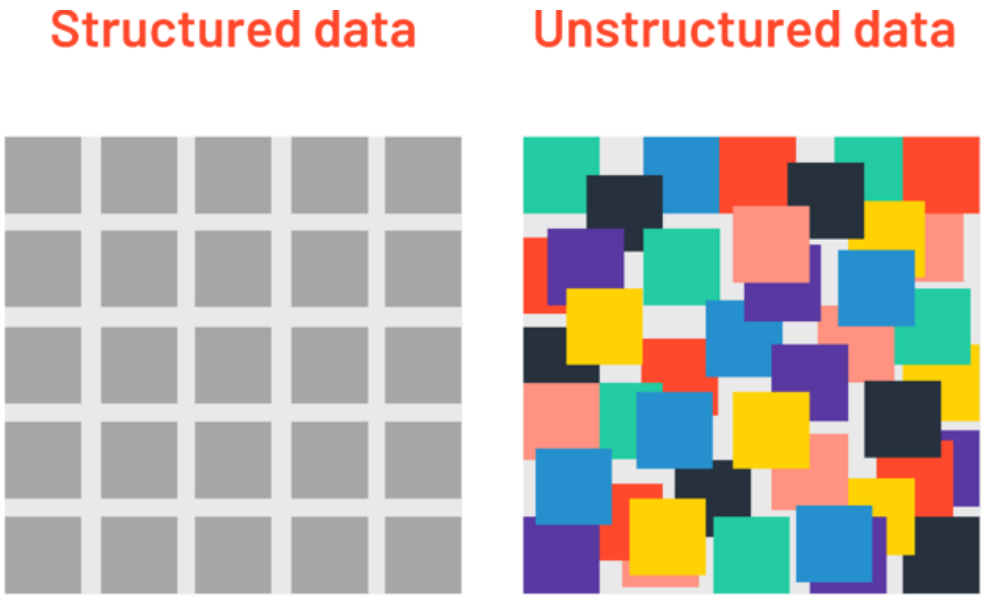
Structure, Unstructured and Semi-structured Data



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Structured Vs Unstructured Data

- **Structured data** is highly-organized and formatted in a way so it's easily searchable in relational databases.
- **Unstructured data** has no pre-defined format or organization, making it much more difficult to collect, process, and analyze.



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Structured Data

- Consider a relation EMPLOYEE

Employee	Manager	Salary
Smith	Jones	50000
Chang	Smith	60000
Ivy	Smith	50000

- This is an example of structured data.

Query:

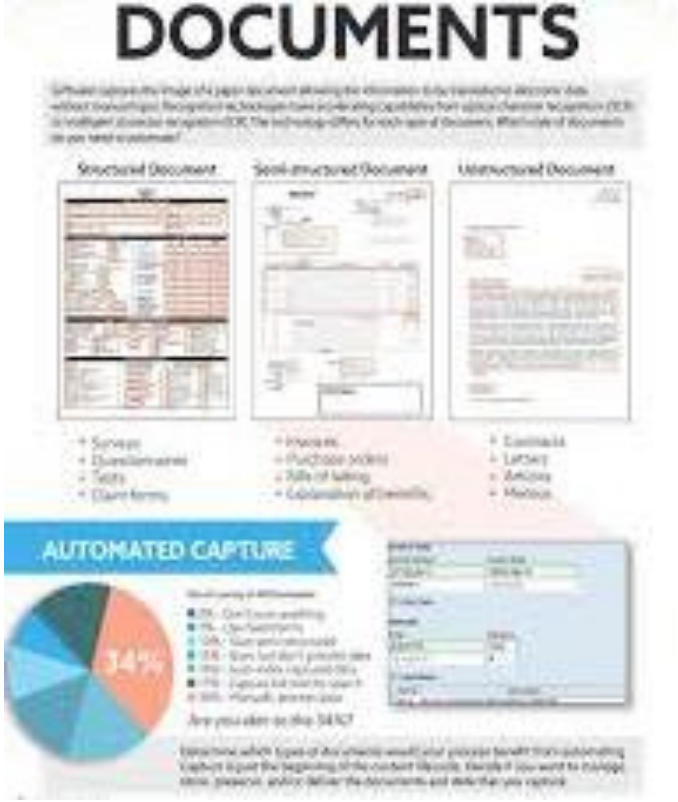
$$Salary < 60000 \ AND \ Manager = Smith$$

Exact Match

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Unstructured Data

- Typically refers to free text
- Allows
  - Keyword queries including operators
  - More sophisticated “concept” queries, for example,
  - *find all web pages dealing with drug abuse*



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Unstructured Data Example

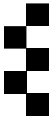
Home Buying Contract Documents

- can vary from city to city, and even county to county
- a contract document in a particular state will have a standard list of clauses
- Other variable information like interest rate for financing, down-payment amount, closing dates, and so on.
- It could also include pictures taken during a home inspection.

Semi-structured Data

- Almost no data is “unstructured”
- **For example**, this slide has distinctly identified zones such as the Title and Bullets
- Facilitates “semi-structured” search such as
  - *Title contains data AND Bullets contain search*
- Or even
  - *Title is about Object Oriented Programming AND Author something like stro*rup*
  - where * is the wild-card operator

Databases and IR Systems: A Comparison



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Structure and Unstructured data

- **Databases** deal with **structured information** retrieval through well-defined formal languages for representation and manipulation based on the theoretically founded data models.
  - Efficient algorithms have been developed for operators that allow rapid execution of complex queries.
- **IR,** on the other hand, deals with **unstructured search** with possibly vague query or search semantics.

Schema

- **Databases** have fixed schemas defined in some data model such as the relational model.
- **IR system** has no fixed data model;
  - IR views data or documents according to some scheme
    - such as the vector space model

Queries

- **Databases** using the relational model **employ SQL** for queries and transactions.
- In **IR systems**, there is **no fixed language** for defining the structure (schema) of the document or for operating on the document
  - queries tend to be a set of query terms (keywords) or a free-form natural language phrase.

Queries' Results

- **A database** query returns a new relation (table) providing an **exact answer** for the current state of the database.
- **An IR query** result is a **list of document id's**, or some pieces of text or multimedia objects (images, videos, and so on), or a list of links to Web pages.

Matching

- The result of a **database query** is an exact answer.
  - if no matching records (tuples) are found in the relation, the result is empty (null).
- On the other hand, the answer to a user request in an **IR query** represents the IR system's best attempt at retrieving the information most relevant to that query.